



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

Reinforcement Learning (AI3000)

Lecture 04

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Utility Function

- A **utility function** assigns value to a reward process
- For an MDP with a finite horizon N , the utility function is defined as

$$u : \mathbb{R}^N \rightarrow \mathbb{R}, \quad u : (R_1, \dots, R_N) \mapsto u(R_1, \dots, R_N).$$

- For an MDP with an infinite horizon, the utility function is defined as

$$u : \mathbb{R}^{\mathbb{N}} \rightarrow \mathbb{R}, \quad (R_1, R_2, \dots) \mapsto u(R_1, R_2, \dots).$$

- The most commonly studied utility functions are **additive**:

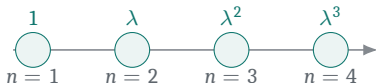
$$\text{Finite horizon :} \quad u(R_1, \dots, R_N) = \sum_{n=1}^N u_n(R_n), \quad \mathbb{E}[u(R_1, \dots, R_N)] = \sum_{n=1}^N \mathbb{E}[u_n(R_n)],$$

$$\text{Infinite horizon :} \quad \mathbb{E}[u(R_1, R_2, \dots)] = \lim_{N \rightarrow \infty} \sum_{k=1}^N \mathbb{E}[u_k(R_k)].$$

Utility Separates Timing and Risk Preferences

$$u_n(R_n) = \underbrace{\lambda^{n-1}}_{\text{timing preference}} \underbrace{f(R_n)}_{\text{risk sensitivity}}, \quad 0 \leq \lambda \leq 1.$$

TIMING PREFERENCE: λ



$\lambda = 1$: time-indifferent

$0 < \lambda < 1$: later rewards receive smaller weights

Smaller λ : see high rewards sooner

RISK SENSITIVITY: f

Strategy A: win 100 with probability 1.

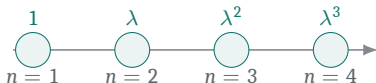
Strategy B: win 0 with probability 0.5;
win 200 with probability 0.5.

Risk attitude	Utility	Strategy preference
Risk-averse	$f(R_n) = \sqrt{R_n}$	prefers A
Risk-neutral	$f(R_n) = R_n$	indifferent
Risk-seeking	$f(R_n) = R_n^2$	prefers B

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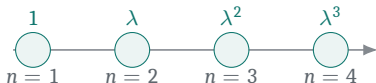
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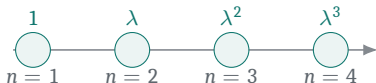
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f captures risk-sensitivity of agent

Optimality Criteria

01

**Expected total reward
criterion**

02

**Expected discounted
reward criterion**

03

**Long-run average
reward criterion**

Unless explicitly stated, we will assume that $\mathcal{S}, \mathcal{A}$ are finite sets.
Furthermore, we will assume that the reward process is bounded, i.e., there exists $M > 0$ such that

$$\mathbb{P}\left(\left\{-M \leq R_n \leq M\right\}\right) = 1 \quad \forall n.$$

Expected Total Reward Criterion

- For a finite horizon MDP with horizon N , the **expected total reward** under policy π is defined as

$$\mathcal{R}_N(\rho, \pi) := \mathbb{E}^{(\rho, \pi)} \left[\sum_{n=1}^N R_n \right] = \sum_{s \in \mathcal{S}} \rho(s) \mathbb{E}^{\pi} \left[\sum_{n=1}^N R_n \mid \{S_0 = s\} \right].$$

- For an infinite horizon MDP, the expected total reward under policy π is defined as

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- The quantities $v_N^{\pi}, v_{\infty}^{\pi}$ are called **policy value functions**

⊙ A Natural Question:

For an infinite-horizon MDP, why is the expected total reward not defined as $\mathcal{R}_{\infty}(\rho, \pi) = \mathbb{E}^{(\rho, \pi)} \left[\sum_{n=1}^{\infty} R_n \right]$?

Answer: Because $\sum_{n=1}^{\infty} R_n$ may not exist!

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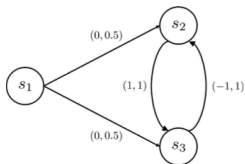
- The quantities v_N^{π} , v_{∞}^{π} are called **policy value functions**

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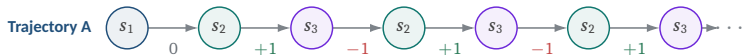
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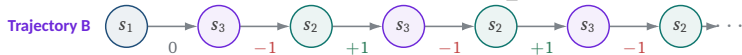
An Infinite Horizon MDP Example



- Say we wish to compute $v_{\infty}^{\pi}(s_1) = \mathbb{E}^{(\rho, \pi)} \left[\sum_{n=1}^{\infty} R_n \mid \{S_0 = s_1\} \right]$
- Starting from $S_0 = s_1$, the first transition is to either s_2 or s_3 ; thereafter, the process alternates between s_2 and s_3



↑ **Trajectory A, probability 1/2:** $\left(\sum_{n=1}^N R_n \right)_{N \geq 1} = (0, 1, 0, 1, \dots)$



↑ **Trajectory B, probability 1/2:** $\left(\sum_{n=1}^N R_n \right)_{N \geq 1} = (0, -1, 0, -1, \dots)$

Pathwise: the partial sums oscillate, so $\sum_{n=1}^{\infty} R_n := \lim_{N \rightarrow \infty} \sum_{n=1}^N R_n$ **does not exist!**

Exercise. Show that $\lim_{N \rightarrow \infty} \mathbb{E}^{(\rho, \pi)} \left[\sum_{n=1}^N R_n \right]$ exists, and compute its value.

Optimal Policy: Expected Total Reward Criterion

Definition (Optimal Policy under Expected Total Reward Criterion)

A policy π^* is said to be **optimal** under the expected total reward criterion if:

1. Finite-horizon setting:

The policy π^* satisfies the condition

$$v_N^{\pi^*}(s) \geq v_N^{\pi}(s) \quad \forall s \in \mathcal{S}, \quad \forall \pi \in \Pi^{\text{HR}}.$$

2. Infinite-horizon setting:

The policy π^* satisfies the condition

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Exercise: Show that

$$\pi^* \text{ optimal} \iff \mathcal{R}_N(\rho, \pi^*) \geq \mathcal{R}_N(\rho, \pi) \quad \forall \rho \in \Delta(\mathcal{S}), \pi \in \Pi^{\text{HR}},$$

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Expected Discounted Reward Criterion

- For a finite horizon MDP with horizon N , the **expected discounted reward** under policy π and discount factor $0 \leq \lambda < 1$ is defined as

$$\mathcal{R}_{N,\lambda}(\rho, \pi) := \mathbb{E}^{(\rho, \pi)} \left[\sum_{n=1}^N \lambda^{n-1} R_n \right] = \sum_{s \in \mathcal{S}} \rho(s) \mathbb{E}^{\pi} \left[\sum_{n=1}^N R_n \mid \{S_0 = s\} \right].$$

- For an infinite horizon MDP, the expected discounted reward under policy π is defined as

$$\mathcal{R}_{\infty,\lambda}(\rho, \pi) := \lim_{N \rightarrow \infty} \mathcal{R}_{N,\lambda}(\rho, \pi) = \lim_{N \rightarrow \infty} \mathbb{E}^{(\rho, \pi)} \left[\sum_{n=1}^N \lambda^{n-1} R_n \right]$$

- If the rewards are bounded, i.e., there exists $M > 0$ such that

$$\mathbb{P}(\{-M \leq R_n \leq M\}) = 1 \quad \forall n,$$

then the limit can be passed inside the summation (courtesy of the dominated convergence theorem) and we get

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A Nice Interpretation of Discounting

- Suppose that $\tau \sim \text{Geometric}(1 - \lambda)$ is independent of everything else
- Then,

$$\begin{aligned} \mathbb{E}^{(\rho, \pi)} \left[\sum_{n=1}^{\tau} R_n \right] &= \sum_{t=1}^{\infty} \sum_{n=1}^t \lambda^{t-1} (1 - \lambda) \mathbb{E}^{(\rho, \pi)} [R_n] = \sum_{n=1}^{\infty} \sum_{t=n}^{\infty} \lambda^{t-1} (1 - \lambda) \mathbb{E}^{(\rho, \pi)} [R_n] \\ &= \sum_{n=1}^{\infty} \lambda^{n-1} \mathbb{E}^{(\rho, \pi)} [R_n] \stackrel{\text{DCT}}{=} \mathbb{E}^{(\rho, \pi)} \left[\sum_{n=1}^{\infty} \lambda^{n-1} R_n \right] = \mathcal{R}_{\infty, \lambda}(\rho, \pi). \end{aligned}$$

- Thus, expected discounted reward framework in infinite horizon MDPs caters to scenarios where a policy stops at a random time that is Geometric and independent of states, actions, and rewards
 - Clinical trials: drug trials until patient dies

Optimal Policy: Discounted Reward Criterion

Definition (Optimal Policy under Expected Discounted Reward Criterion)

A policy π^* is said to be **optimal** under the expected discounted reward criterion with discount factor $0 \leq \lambda < 1$ if:

1. Finite-horizon setting:

The policy π^* satisfies the condition

$$v_{N,\lambda}^{\pi^*}(s) \geq v_{N,\lambda}^{\pi}(s) \quad \forall s \in \mathcal{S}, \quad \forall \pi \in \Pi^{\text{HR}}.$$

2. Infinite-horizon setting:

The policy π^* satisfies the condition

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Exercise: Show that

$$\pi^* \text{ optimal} \iff \mathcal{R}_{N,\lambda}(\rho, \pi^*) \geq \mathcal{R}_{N,\lambda}(\rho, \pi) \quad \forall \rho \in \Delta(\mathcal{S}), \pi \in \Pi^{\text{HR}},$$

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Long-Run Average Reward Criterion

- Applicable to infinite horizon MDPs where the interest is in long-run behaviour of $\{(S_n, A_n)\}_{n \geq 0}$ and $\{R_n\}_{n \geq 1}$
- For an infinite-horizon MDP, the long-run average reward under a policy π is defined as

$$\mathcal{G}(\rho, \pi) := \lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E}^{(\rho, \pi)} \left[\sum_{n=1}^N R_n \right] = .$$

- The limits above may not always exist
 - If the limits do not exist, then we replace them with $\lim_{N \rightarrow \infty}$ with $\limsup_{N \rightarrow \infty}$ or $\liminf_{N \rightarrow \infty}$
 - If π is a stationary policy, then the above limits always exist
- If rewards are bounded, the limits can be passed inside the expectation (dominated convergence theorem)

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Optimal Policy: Long-Run Average Reward Criterion

Definition (Optimal Policy under Long-Run Average Reward Criterion)

A policy π^* is said to be **optimal** under the long-run average reward criterion if:

$$g^{\pi^*}(s) \geq g^{\pi}(s) \quad \forall s \in \mathcal{S}, \forall \pi \in \Pi^{\text{HR}}.$$

Exercise: Show that

$$\pi^* \text{ optimal} \quad \Longleftrightarrow \quad \mathcal{G}(\rho, \pi^*) \geq \mathcal{G}(\rho, \pi) \quad \forall \rho \in \Delta(\mathcal{S}), \pi \in \Pi^{\text{HR}}.$$

The Three Optimality Criteria in a Nutshell

Criterion	Horizon	Objective	Policy value function	Policy optimality
Expected total reward	Finite	$\mathcal{R}_N(\rho, \pi) = \mathbb{E}^{(\rho, \pi)} \left[\sum_{n=1}^N R_n \right]$	$v_N^\pi(s) = \mathbb{E}^\pi \left[\sum_{n=1}^N R_n \mid S_0 = s \right]$	$v_N^{\pi^*}(s) \geq v_N^\pi(s)$
	Infinite	$\mathcal{R}_\infty(\rho, \pi) = \lim_{N \rightarrow \infty} \mathbb{E}^{(\rho, \pi)} \left[\sum_{n=1}^N R_n \right]$	$v_\infty^\pi(s) = \lim_{N \rightarrow \infty} \mathbb{E}^\pi \left[\sum_{n=1}^N R_n \mid S_0 = s \right]$	$v_\infty^{\pi^*}(s) \geq v_\infty^\pi(s)$
Expected discounted reward	Finite	$\mathcal{R}_{N, \lambda}(\rho, \pi) = \mathbb{E}^{(\rho, \pi)} \left[\sum_{n=1}^N \lambda^{n-1} R_n \right]$	$v_{N, \lambda}^\pi(s) = \mathbb{E}^\pi \left[\sum_{n=1}^N \lambda^{n-1} R_n \mid S_0 = s \right]$	$v_{N, \lambda}^{\pi^*}(s) \geq v_{N, \lambda}^\pi(s)$
	Infinite	$\mathcal{R}_{\infty, \lambda}(\rho, \pi) = \mathbb{E}^{(\rho, \pi)} \left[\sum_{n=1}^{\infty} \lambda^{n-1} R_n \right]$	$v_{\infty, \lambda}^\pi(s) = \mathbb{E}^\pi \left[\sum_{n=1}^{\infty} \lambda^{n-1} R_n \mid S_0 = s \right]$	$v_{\infty, \lambda}^{\pi^*}(s) \geq v_{\infty, \lambda}^\pi(s)$
Long-run average reward	Infinite	$\mathcal{G}(\rho, \pi) = \lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E}^{(\rho, \pi)} \left[\sum_{n=1}^N R_n \right]$	$g^\pi(s) = \lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E}^\pi \left[\sum_{n=1}^N R_n \mid S_0 = s \right]$	$g^{\pi^*}(s) \geq g^\pi(s)$

Here $0 \leq \lambda < 1$. Each optimality inequality holds for every $s \in \mathcal{S}$ and every $\pi \in \Pi^{\text{HR}}$.